1.1.1. Calculate Momentum

Write a program that accepts the mass of an object (in kilograms) and its velocity (in meters per second), then calculates and displays the momentum of the object. The momentum p is calculated using the formula:

$$p = m \times v$$

where:
m is the mass of the object (in kilograms).
v is the velocity of the object (in meters per second).

Input Format:

- A single floating-point number representing the mass of the object in kilograms.
- A single floating-point number representing the velocity of the object in meters per second.

Output Format:

- The output will display calculated momentum with appropriate units (kgm/s) (rounded up to 2 decimal places).

Sample Test Cases +

calculate...

```python
#Type Content here...
m = float(input())
v = float(input())
p = m * v

print('%.2f' % p, end="")
print('kgm/s')
```

Average time 0.017 s 16.50 ms
Maximum time 0.024 s 24.00 ms

2 out of 2 shown test case(s) p
2 out of 2 hidden test case(s) p

Test case 1 24 ms Debug

Expected output	Actual output
5.0	5.0
10.0	10.0
50.00kgm/s	50.00kgm/s

Test case 2 18 ms

Terminal Test cases

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  - 1.1. Practice Lab Assignment
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- 4. Practical 4
- 5. Practical 5

1.1.2. Conditional Calculation Based on the Number o... 00:15

Write a Python program that accepts an integer n as input. Depending on the number of digits in n.

Constraints:
$1 \leq n \leq 999$

Input Format:
- The input consists of a single integer n.

Output Format:
- If n is a single-digit number, print its square.
- If n is a two-digit number, print its square root (rounded to two decimal places).
- If n is a three-digit number, print its cube root (rounded to two decimal places).
- Else print "Invalid".

Sample Test Cases +

condition... Submit

```python
#write your code here...
import math

n = int(input())

if 1 <= n <= 999:
    if 1 <= n <= 9:
        print(n * n)
    elif 10 <= n <= 99:
        print(f"{math.sqrt(n):.2f}")
    else:
```

Average time 0.005 s 5.00 ms | Maximum time 0.006 s 6.00 ms

4 out of 4 shown test case(s) p
3 out of 3 hidden test case(s) p

Test case 1 5 ms Debug
Expected output: 9 81
Actual output: 9 81

Test case 2 6 ms

Test case 3 4 ms

Terminal | Test cases

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1.1. Practice Lab Assignment

1.1.1. Calculate Momentum

1.1.2. Conditional Calculation Based on the...

1.1.3. Days between Two Dates

1.1.4. Reverse a Number

1.1.5. Multiplication Table

1.2. Lab Assignment

2. Practical 2

3. Practical 3

4. Practical 4

5. Practical 5

1.1.3. Days between Two Dates 00:35

Write a Python program to calculate number of days between two dates.

Input Format:

- The first line contains the first date in the format $YYYY - MM - DD$.
- The second line contains the second date in the format $YYYY - MM - DD$.

Output Format:

- An integer representing the number of days between the given dates.

Note:

- The first date should always be considered earlier than the second date.

Sample Test Cases +

Explorer noOfDay... Submit Debugger

```
"""from datetime import date

#write your code here..."""
from datetime import date
from datetime import datetime

# Read two dates as input
date1 = input()
date2 = input()

# Convert string to datetime object
```

Average time 0.018 s 18.00 ms

Maximum time 0.039 s 39.00 ms

2 out of 2 shown test case(s) p

2 out of 2 hidden test case(s) p

Test case 1 39 ms Debug

Expected output	Actual output
2014-07-02	2014-07-02
2014-11-02	2014-11-02
123	123

Test case 2 12 ms

Terminal Test cases

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- 2. Practical 2
- 3. Practical 3
- 4. Practical 4
- 5. Practical 5

1.1.4. Reverse a Number 00:13

Write a Python program to reverse the digits of the number and print the reversed number.

Input Format:

- The input is a single integer.

Output Format:

- The output prints a single integer, which is the reversed number.

Sample Test Cases +

Explorer

reverseN... Submit

```
#write your code here...
num = int(input().strip())
reversed_num = int(str(num)[::-1])
print(reversed_num)
```

Debugger

Average time	Maximum time
0.009 s	0.012 s
9.20 ms	12.00 ms

2 out of 2 shown test case(s) p

3 out of 3 hidden test case(s) p

Test case 1 12 ms Debug

Expected output	Actual output
5367	5367
7635	7635

Test case 2 8 ms

Terminal Test cases

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1.1.5. Multiplication Table

Write a Python program that takes an integer as input and prints the multiplication table for that integer from 1 to 10.

Input Format:

- The first line of input contains an integer that represents the number for which the multiplication table is to be printed.

Output Format:

- Print the multiplication table for the given number in the format:

```
<number> x <multiplier> = <result>
```

Note:

- The character "x" is used in the output to represent multiplication.
- Refer to the visible test-cases for more clarity.

Sample Test Cases +

multiplica...

```python
#write your code here...
n = int(input())
for i in range(1, 11):
    print(f"{n} x {i} = {n*i}")
```

Average time 0.006 s 6.00 ms | Maximum time 0.007 s 7.00 ms

2 out of 2 shown test case(s) p
2 out of 2 hidden test case(s) p

Test case 1 5 ms Debug

Expected output	Actual output
8	8
8 x 1 = 8	8 x 1 = 8
8 x 2 = 16	8 x 2 = 16
8 x 3 = 24	8 x 3 = 24
8 x 4 = 32	8 x 4 = 32
8 x 5 = 40	8 x 5 = 40

Terminal Test cases

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1.2.1. Pass or Fail

00:17

Write a Python program that accepts the number of courses and the marks of a student in those courses.

The grade is determined based on the aggregate percentage:

- If the aggregate percentage is greater than 75, the grade is Distinction.
- If the aggregate percentage is greater than or equal to 60 but less than 75, the grade is First Division.
- If the aggregate percentage is greater than or equal to 50 but less than 60, the grade is Second Division.
- If the aggregate percentage is greater than or equal to 40 but less than 50, the grade is Third Division.

Input Format:

- The first input will be an integer n, the number of courses.
- The second input will be n integers representing the marks of the student in each of the n courses, separated by a space.

Output Format:

If the student passes all courses:

- Print the aggregate percentage (formatted to two decimal places).
- Print the grade based on the aggregate percentage.

If the student fails any course (marks < 40 in any course), print:

- "Fail".

Note:

- Aggregate Percentage: Average performance across all courses

Explorer

passorFa...

Submit

```python
# write your code here
def calculate_grade(num_courses, marks):
    if len(marks) != num_courses:
        return "Error: Number of marks does not match number of courses."

    if any(mark < 40 for mark in marks):
        return "Fail"

    aggregate_percentage = sum(marks) / num_courses

    if aggregate_percentage > 75:
        grade = "Distinction"
    elif aggregate_percentage > 60:
        grade = "First Division"
    elif aggregate_percentage > 50:
        grade = "Second Division"
    elif aggregate_percentage >= 40:
        grade = "Third Division"
    else:
        grade = "Fail"

    return f"Aggregate Percentage: {aggregate_percentage:.2f}\nGrade: {grade}"
```

Debugger

Terminal | Test cases

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1.2.1. Pass or Fail

Write a Python program that accepts the number of courses and the marks of a student in those courses.

The grade is determined based on the aggregate percentage:

- If the aggregate percentage is greater than 75, the grade is Distinction.
- If the aggregate percentage is greater than or equal to 60 but less than 75, the grade is First Division.
- If the aggregate percentage is greater than or equal to 50 but less than 60, the grade is Second Division.
- If the aggregate percentage is greater than or equal to 40 but less than 50, the grade is Third Division.

Input Format:

- The first input will be an integer n, the number of courses.
- The second input will be n integers representing the marks of the student in each of the n courses, separated by a space.

Output Format:

If the student passes all courses:

- Print the aggregate percentage (formatted to two decimal places).
- Print the grade based on the aggregate percentage.

If the student fails any course (marks < 40 in any course), print:

- "Fail".

Note:

- Aggregate Percentage: Average performance across all courses

passorFa... Submit

```python
# write your code here
def calculate_grade(num_courses, marks):
    if len(marks) != num_courses:
        return "Error: Number of marks does not match number of courses."

    if any(mark < 40 for mark in marks):
        return "Fail"

    aggregate_percentage = sum(marks) / num_courses

```

Average time 0.023 s 22.80 ms | Maximum time 0.050 s 50.00 ms

5 out of 5 shown test case(s) p
5 out of 5 hidden test case(s) p

Test case 1 21 ms Debug

Expected output	Actual output
4	4
55 65 78 89	55 65 78 89
Aggregate Percentage: 71.75	Aggregate Percentage: 71.75
Grade: First Division	Grade: First Division

Test case 2

Terminal Test cases

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Search course ctrl + k

1.2.2. Fibonacci Series 00:35

Write a Python program that uses recursion to print the first n terms of the Fibonacci series.

Input Format:

- A single integer n representing the number of terms to generate.

Output Format:

- A single line containing the first n terms of the Fibonacci sequence, separated by spaces.

Sample Test Cases +

Explorer | fibonacci.... | Submit | Debugger

```
"""def fibonacci(n):

    # write the code..

        return fibonacci(   )


n = int(input())
for i in range(1, n + 1):
    print(fibonacci(i), end=" ")
"""
```

Average time	Maximum time
0.019 s	0.053 s
19.50 ms	53.00 ms

2 out of 2 shown test case(s) p

2 out of 2 hidden test case(s) p

Test case 1 10 ms Debug

Expected output	Actual output
5	5
0 1 1 2 3	0 1 1 2 3

Test case 2 5 ms

Terminal Test cases

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1.2.3. Pattern - 1

00:16

Write a Python program to print a pattern of asterisks in the form of a right-angled triangle.

Input Format:

- The input is an integer, representing the number of rows in the pattern.

Output Format

- The output should display the pattern of asterisks (*), with each row containing an increasing number of asterisks.

Note: Refer to the displayed test cases for the sample pattern.

Sample Test Cases +

Explorer rightangl... Submit Debugger

```python
#Type Content here...
n = int(input())
for i in range(1, n + 1):
    for j in range(i):
        print("*", end=" ")
    print()
```

Average time	Maximum time
0.006 s	0.007 s
6.33 ms	7.00 ms

2 out of 2 shown test case(s) p
4 out of 4 hidden test case(s) p

Test case 1 7 ms Debug

Expected output	Actual output
5	5
*	*
* *	* *
* * *	* * *
* * * *	* * * *
* * * * *	* * * * *

Terminal Test cases

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